# SDR-Based Spectrum Monitoring System for FM Broadcast Compliance Assessment

## Design Considerations for SDR-Based Monitoring Architectures

- *Reconfigurability and Flexibility:* The SDR platform must exhibit inherent reconfigurability to dynamically adjust signal processing parameters during monitoring operations. Since core functions—demodulation, detection thresholding, channel occupancy analysis, and compliance reporting—are virtualized in software rather than fixed in hardware, the system can adapt to evolving regulatory requirements, expand to adjacent frequency bands, or integrate advanced detection techniques (e.g., cyclostationary feature analysis, machine learning-based classification) via remote software updates without hardware modification.

- *Cost-Effectiveness:* By leveraging the HackRF One with embedded processors (DragonBoard 410c, Raspberry Pi), the proposed architecture achieves persistent spectrum monitoring at substantially reduced cost versus traditional spectrum analyzers ($10,000–$50,000 USD per channel). With a target deployed cost of $\leq$$1000 USD per node—encompassing SDR, processor, RF front-end, enclosure, and deployment—the design enables order-of-magnitude scalability for distributed regulatory monitoring networks.

- *Remote Monitoring Capability:* The SDR-based system must transmit captured RF signal metadata, demodulated content, and compliance measurement results to a centralized server via IP-based networks, enabling regulatory personnel to monitor multiple FM broadcast stations from a central operations center. All data-in-transit must be protected using authenticated encryption (TLS 1.3 or higher, or IPsec-based VPN tunnels with AES-256-GCM). Access to the web-based monitoring interface shall be restricted to authenticated and authorized personnel via a role-based access control (RBAC) mechanism, with audit logging of all administrative actions.

- *Multi-Station Monitoring:* The digital signal processing (DSP) pipeline shall support concurrent real-time demodulation and compliance analysis of a minimum of two FM broadcast channels on the target embedded platform. The system architecture shall be scalable to accommodate additional channels contingent upon available computational resources. Processing capacity shall be validated through empirical benchmarking on the target hardware prior to deployment, with the maximum sustainable concurrent channel count documented in the System Performance Specification.

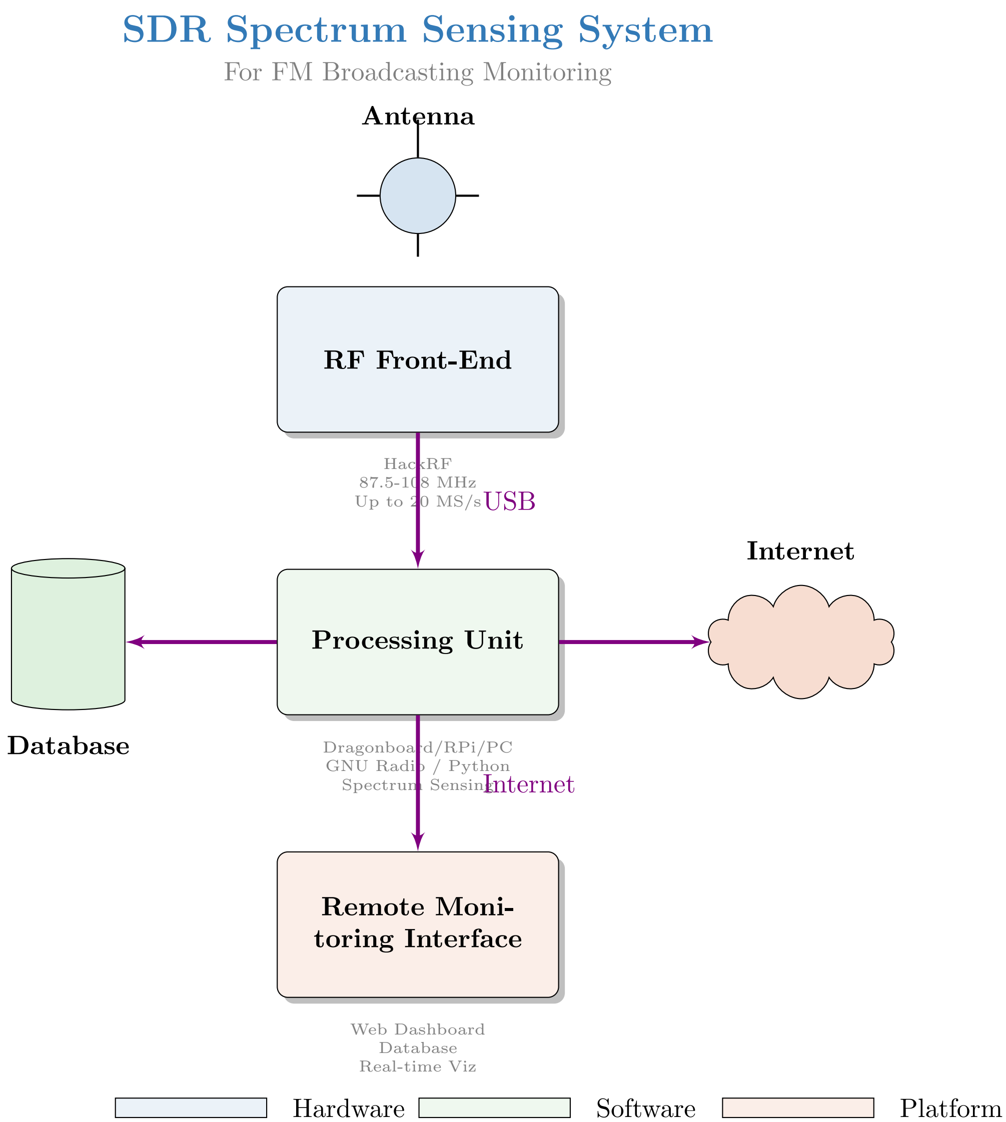


Figure 1: Core Architectural Design

### Core Architectural Components

- *RF Front-End (Hardware Layer):* At the physical layer, a low-cost SDR receiver—the HackRF One—serves as the antenna-coupled RF front end. Its primary functions include: capturing incident RF energy in the VHF-II band (87.5–108 MHz), down-converting the signal to complex baseband via a heterodyne-conversion architecture, and digitizing it using a 8-bit ADC at up to 20 MS/s via USB 2.0. Instead of a direct-conversion architecture, the HackRF One uses a MAX2837 RF transceiver, which performs a two-stage heterodyne conversion it is not a zero-IF direct-conversion device. This matters for the compliance monitoring application because direct-conversion receivers exhibit DC offset and IQ imbalance artifacts that degrade near-DC spectral measurements However, sustained real-world throughput for the HackRF at the maximum rate is often lower than the theoretical limit due to USB 2.0 overhead and host buffering; practical deployments frequently cap at 1015 MS/s without drops.

Critical performance parameters at this stage include instantaneous bandwidth, dynamic range, noise figure, and sampling rate, which collectively determine the signal integrity and measurement accuracy achievable for FM broadcast monitoring. A broadband antenna optimized for the VHF-II band couples to the SDR input, and a low-noise amplifier (LNA) may be integrated into the receive chain to improve system sensitivity and compensate for feedline attenuation in low-signal environments.

- *Signal Acquisition and Digitization:* The SDR hardware streams raw complex baseband (I/Q) samples to the embedded host processor via a USB 2.0 high-speed interface. These quadrature samples represent the receiver's full instantaneous bandwidth—up to 20 MHz for the HackRF One—enabling capture of multiple FM broadcast channels (200 kHz spacing) within a single acquisition window. This wideband architecture provides a critical advantage over traditional narrowband receivers: rather than sequentially tuning to individual carriers, the system can simultaneously demodulate and analyze all FM stations falling within the instantaneous bandwidth, significantly reducing scan latency and enabling real-time correlation of spectral events across channels.

- *Data Management and Reporting Layer:* Processed measurement results are persisted to a time-series database backend, structured around time-stamped records with metadata tags (frequency, location, station ID, compliance status). This schema enables efficient historical trending, statistical compliance analysis, and event-triggered alerting. A RESTful API and web-based dashboard provide remote access for regulatory personnel to query measurement data, visualize real-time spectrograms and spectral occupancy heatmaps, and generate audit-ready compliance reports aligned with regulatory templates (e.g., ITU-R SM.1880, FCC Part 73 technical rules). Automated alert mechanisms—configurable via threshold policies (absolute, statistical, or ML-based anomaly detection)—notify engineers via email, SMS, or webhook integrations when violations are detected, with escalation policies for critical events.

- *Software Processing Engine:* The processing engine implements the end-to-end digital signal processing (DSP) pipeline for *FM broadcast compliance monitoring.* Unlike communication-centric SDR applications, its primary output comprises *auditable measurements, evidentiary records, and compliance reports.*

  The six-layer compliance monitoring pipeline is developed, embracing the following stages:

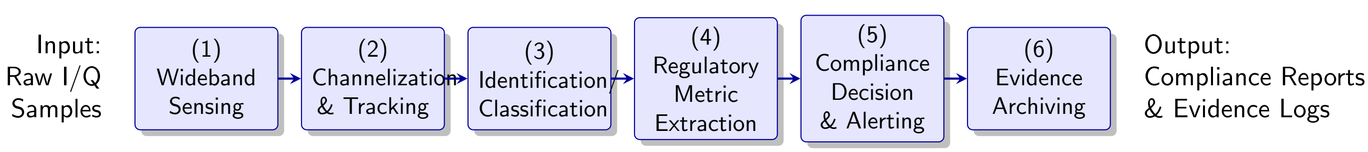


Figure 2: SDR-based FM broadcast compliance monitoring processing pipeline. Raw I/Q samples flow through six sequential stages from wideband spectrum sensing to evidence archiving, enabling automated regulatory compliance assessment.

**Wideband Spectrum Sensing (Occupancy and Discovery).** This stage estimates the Power Spectral Density (PSD) of received power across frequency to (i) discover active FM emissions, (ii) maintain a time-stamped occupancy map of the band, and (iii) provide measurement features that feed station tracking and compliance checks. Key outputs include detected station candidates (center frequency estimates), received signal strength indicators (RSSI), and activity timelines.

1. *Controlled Data Acquisition.* The SDR compliance monitoring architecture is characterized by time-critical data ingress from the peripheral hardware to the host processor via a USB interface. Discrepancies between the continuous sampling rate and downstream processing or storage throughput can result in buffer overflows, excessive latency accumulation, or unbounded resource consumption. Consequently, the integration of a flow control mechanism—specifically backpressure propagation—is essential to regulate data flow, thereby ensuring system stability and deterministic operational behavior.

    > **Experimental Set-up**
    > The validated sensors are placed in close proximity to a common FM reference signal source, ensuring all HackRF One units are configured with identical settings and sample rates, to record data under identical environmental and operational conditions.
    >
    > ▷ All processing should be implemented in the software engine (GNU Radio/Python), ensuring compatibility with embedded targets (Dragonboard 410C, Raspberry Pi).
    >
    > ▷ Benchmark the full pipeline on the target embedded hardware to verify that calibration overhead does not compromise the real-time monitoring requirement of $\geq 2$ simultaneous FM channels.
    >
    > *Recommendation:* All hardware configuration parameters (gain, sample rate, antenna type) are documented to ensure reproducibility. The calibration procedure is executable as an automated script, triggered on-demand or on a scheduled basis, with results pushed to the central database.

2. *Preprocessing.* The fidelity of the acquired measurement data is constrained by hardware impairments inherent to the HackRF One platform when used with the MAX2837 RF transceiver, which implements a two-stage heterodyne downconversion rather than a zero-IF direct-conversion path. Consequently, DC offsets, LO leakage, and I/Q imbalance manifest differently than in direct-conversion SDRs and must be addressed in the digital domain with awareness of the heterodyne architecture.

**Experimental Set-up:** For real captures, the best hardware-aware workflow for HackRF One with MAX2837 is usually: tune slightly off-center, digitally shift if needed, and apply a time-domain DC blocker / mean removal on IQ,

- ▷ **I/Q correction.** I/Q distortion in a heterodyne-based chain typically shows up as mirrored or ghost content around DC, and can also manifest as image lines near the downconverted IF. Implement per-channel I/Q imbalance calibration and compensation using reference signals, updating calibration as LO settings or center frequency change.
- ▷ **DC/LO leakage and center-spike correction.** In a two-stage downconversion path, DC-like spikes and LO leakage can appear at baseband (0 Hz) and at fixed offsets related to the IF downconversion stages. These artifacts are not guaranteed to be fixed at 0 Hz across all tunings; their amplitude and location can vary with tuning. Apply a time-domain DC blocker and a dynamic, channel-specific DC notch near 0 Hz, complemented by LO-leakage calibration to suppress persistent spurs.
- ▷ **Comb-like spikes.** Comb-like, evenly spaced spikes can arise from sampling clock-/quantization artifacts or external intermodulation, and may be exacerbated by I/Q imbalance or LO spurs. Use clock-aware filtering with notches at known clock-harmonic offsets and robust spectral preprocessing to distinguish genuine emissions from artifacts. Maintain a log of identified spurs for traceability.

3. *Spectral Parameter Estimation.* Given a Power Spectral Density (PSD) estimate $\widehat{S}_x(f)$ over a band of interest, derive the following band-level features:

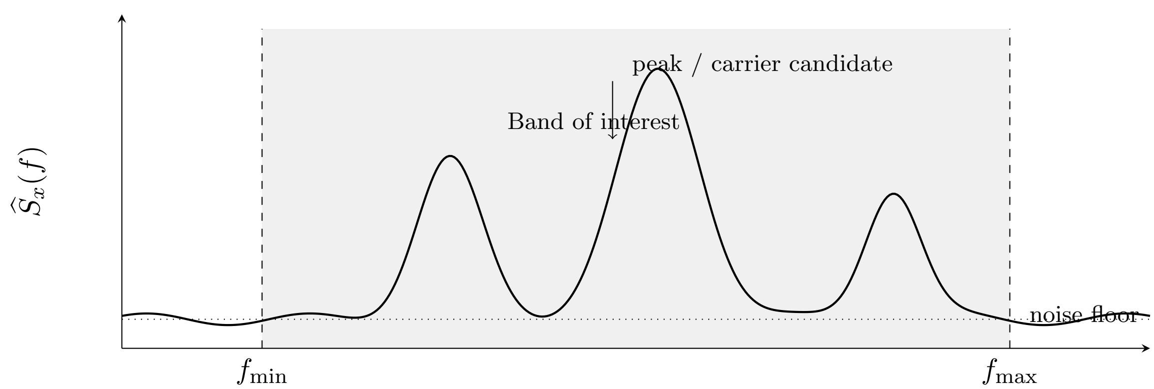


Figure 3: Illustration of a PSD estimate $\widehat{S}_x(f)$ of the received wideband complex I/Q over a frequency band of interest $[f_{\min}, f_{\max}]$, showing a noise floor and multiple detected emissions.

**Experimental Set-up**

- ▷ *Noise Floor Estimation*
- ▷ *Power at specific frequencies:* $\widehat{S}_x(f_k)$ for selected $f_k$ (e.g., nominal channel centers).
- ▷ *Peak frequency:* $f_{\text{pk}} \triangleq \arg\max_f \widehat{S}_x(f)$ (candidate carrier location).
- ▷ *Occupied bandwidth:* $B_{\text{occ}}$ defined as the smallest bandwidth around a center $f_c$ that contains a prescribed fraction of in-band power (e.g., 99%), or equivalently via a thresholded support set.
- ▷ *Center frequency (spectral centroid):* $f_{\text{cent}}$ interpreted as a power-weighted mean frequency over the analyzed band.
- ▷ *SNR proxy:* Robust estimate of in-band signal power relative to a noise-floor estimate, producing $\widehat{\text{SNR}}$ for detection confidence and alert thresholds.

4. *Occupancy Detection and Band Discovery.* Convert $\widehat{S}_x(f)$ into a binary (or probabilistic) occupancy map by comparing against a noise-floor estimate $\widehat{N}(f)$:

$$\mathcal{O}(f) = \mathbf{1}\{\widehat{S}_x(f) > \widehat{N}(f) + \gamma\},$$

where $\gamma$ is a configurable margin (in dB). Connected occupied regions define candidate emissions; their centers, peak locations, and widths provide initial station candidates for downstream channelization and tracking.

**Experimental Set-up:** compliance-oriented and signal-quality features are extracted from each sensor's normalized data:

- ▷ *Standard spectral features:* Spectral flatness, SNR proxy, occupied bandwidth ($B_{\text{occ}}$), carrier frequency error/drift, spectral centroid ($f_{\text{cent}}$)

$$f_{\text{cent}} \triangleq \int f\, \widehat{S}_x(f)\, df \Big/ \int \widehat{S}_x(f)\, df,$$

*Power at specific frequencies Peak frequency Occupied bandwidth Center frequency (spectral centroid)*

- ▷ *FM-specific features:* 19 kHz pilot tone SNR, MPX spectral structure integrity (presence of 38 kHz stereo subcarrier, 57 kHz RDS), deviation profile consistency. *SNR proxy:* estimate in-band signal power relative to a noise-floor estimate (median/robust fit outside detected carriers), producing $\widehat{\text{SNR}}$ for detection confidence and alert thresholds.

*Recommendation:* Prioritize features "Compliance Feature Extraction" module to ensure regulatory relevance. Standardize all features (zero-mean, unit-variance) before downstream similarity computations.

5. *Quantitative Reconstruction of PSD.* When the PSD is degraded by noise, short observation windows, or coarse resolution, apply reconstruction techniques to improve the stability of the

occupancy map and derived features. Examples include: (i) smoothing or averaging across time, (ii) interpolation across frequency bins, and (iii) model-based fitting for the noise floor and/or emission shapes. This stage must preserve auditability by logging parameters and reconstruction settings used for each reported estimate.

6. *Demodulation-Assisted Verification & Station Identification.* After candidate emissions are discovered from $\widehat{S}_x(f)$, each candidate channel is *isolated and decimated* (channelization $\rightarrow$ narrowband I/Q) and a lightweight FM demodulation stage is applied to:

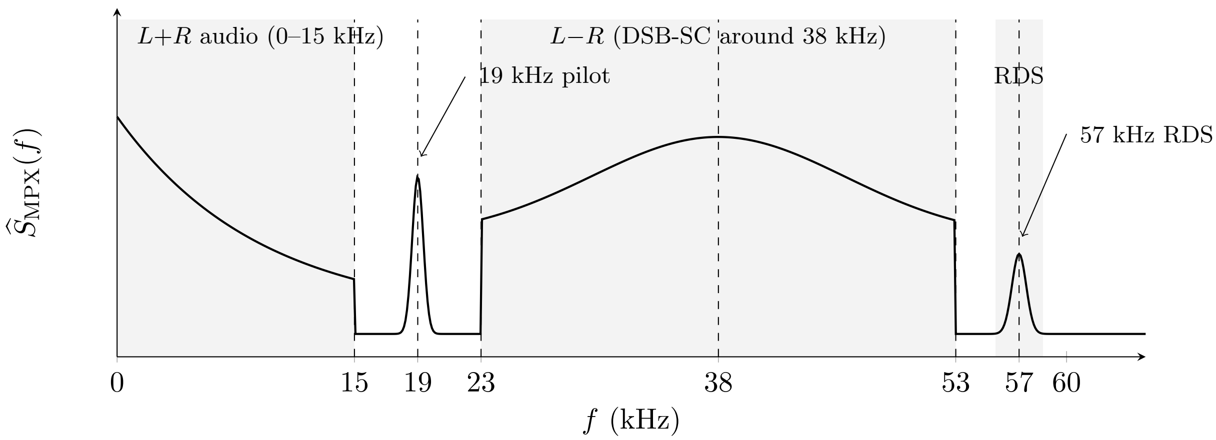


Figure 4: Demodulated baseband spectrum (post-demod) of the audio/composite (MPX) after FM discriminator, centered at 0 Hz audio, with features like 19 kHz pilot, 38 kHz stereo subcarrier, and possibly 57 kHz RDS.

- *Confirm emission type:* Verify that the detected occupancy corresponds to an FM broadcast-like signal (reject interferers and non-broadcast emissions).
- *Extract identification cues:* Detect the 19 kHz stereo pilot and optionally decode Radio Data System (RDS) to obtain station metadata (e.g., PI, PS, PTY when available).
- *Support evidence generation:* Produce short, time-stamped *audio snippets* and/or *RDS logs* linked to occupancy events for audits and investigations.
- *Enable demodulation-derived metrics (optional):* Compute modulation or deviation-related indicators when required by the compliance rule set.